



UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO

FACULTAD DE INGENIERÍA

DIVISIÓN DE INGENIERÍA ELÉCTRICA

INGENIERÍA EN COMPUTACIÓN

LABORATORIO DE COMPUTACIÓN GRÁFICA e
INTERACCIÓN HUMANO COMPUTADORA



REPORTE DE PRÁCTICA N° 04

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GRUPO DE LABORATORIO: 03

GRUPO DE TEORÍA: 06

SEMESTRE 2026 - 2

FECHA DE ENTREGA LÍMITE: 17 de marzo

CALIFICACIÓN: _____

REPORTE DE PRÁCTICA:

1.- Ejecución de los ejercicios que se dejaron, comentar cada uno y capturas de pantalla de bloques de código generados y de ejecución del programa.

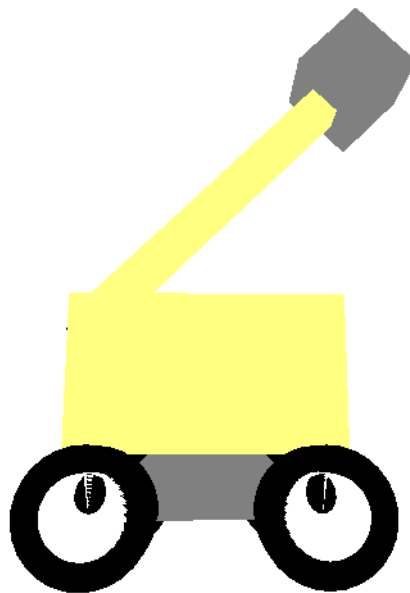
1.- Terminar la Grúa con:

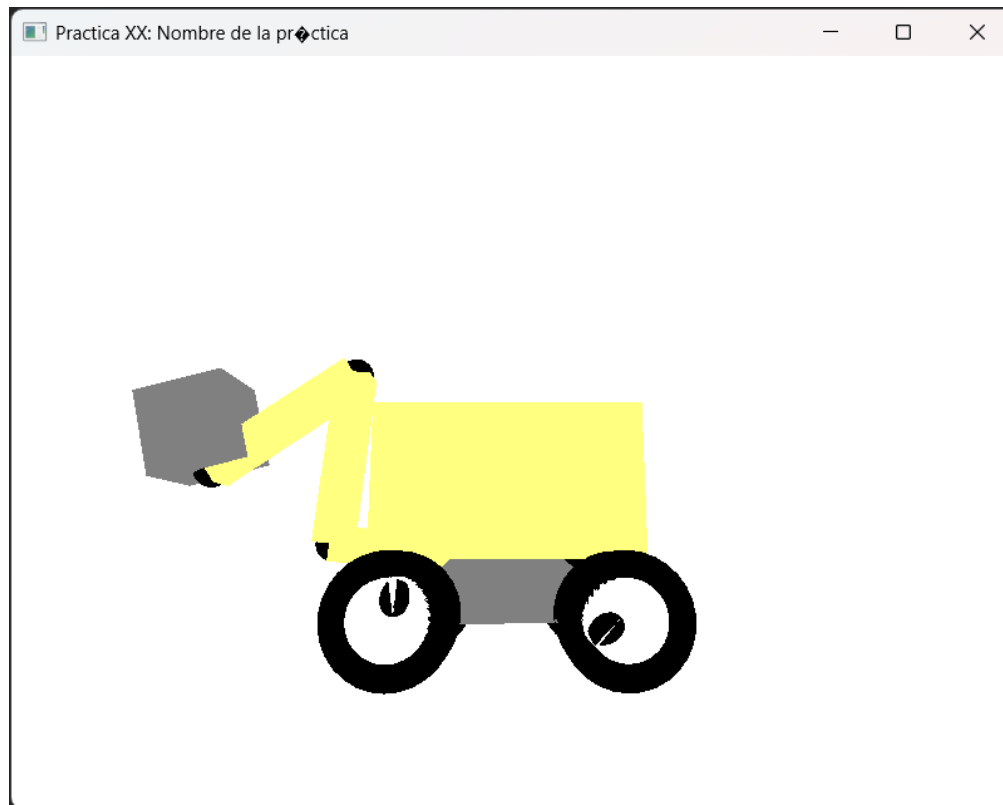
-cuerpo(prisma rectangular)

-base (pirámide cuadrangular)

-4 llantas(4 cilindros) con teclado se pueden girar cada una de las 4 llantas por separado

Capturas de ejecución:





Video de ejecución: <https://www.youtube.com/watch?v=CXf0fFKTAn4>

Código de ejercicio:

```
346 //Inicio Grua-----
347
348 //CREANDO LA CABINA
349 model = glm::mat4(1.0);
350 model = glm::translate(model, glm::vec3(0.0f, 7.0f, -4.0f));
351 //aquí está la jerarquía
352 modelaux = model;
353 model = glm::scale(model, glm::vec3(7.0f, 4.0f, 5.0f));
354 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
355 //la línea de proyección solo se manda una vez a menos que en tiempo de ejecución
356 //se programe cambio entre proyección ortogonal y perspectiva
357 glUniformMatrix4fv(uniformProjection, 1, GL_FALSE, glm::value_ptr(projection));
358 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
359 color = glm::vec3(1.0f, 1.0f, 0.5f); // GRIS
360 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
361 meshList[0]->RenderMesh();
```

```

363 //Base
364 model = modelaux;
365 model = glm::translate(model, glm::vec3(0.0f, -2.0f, 0.0f));
366 modelaux = model;
367 model = glm::scale(model, glm::vec3(6.0f, 3.0f, 6.0f));
368 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
369 color = glm::vec3(0.5f, 0.5f, 0.5f);
370 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
371 meshList[4]->RenderMeshGeometry();
372
373 //Llanta 1
374 model = modelaux;
375 model = glm::rotate(model, glm::radians(90.0f), glm::vec3(0.0f, 0.0f, 1.0f));
376 model = glm::rotate(model, glm::radians(90.0f), glm::vec3(1.0f, 0.0f, 0.0f));
377 model = glm::translate(model, glm::vec3(-1.3f, 3.65f, 2.6f));
378 model = glm::scale(model, glm::vec3(1.5f, 1.5f, 1.5f));
379 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion5()), glm::vec3(0.0f, 1.0f, 0.0f));
380 glm::mat4 modelRin1 = model;
381 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
382 color = glm::vec3(0.0f, 0.0f, 0.0f);
383 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
384 meshList[5]->RenderMeshGeometry();
385
386 //Centro de la llanta
387 model = modelRin1;
388 model = glm::scale(model, glm::vec3(0.6f, 1.05f, 0.6f));
389 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
390 color = glm::vec3(1.0f, 1.0f, 1.0f);
391 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
392 meshList[5]->RenderMeshGeometry();

```

```

394 //Llanta 2
395 model = modelaux;
396 model = glm::rotate(model, glm::radians(90.0f), glm::vec3(0.0f, 0.0f, 1.0f));
397 model = glm::rotate(model, glm::radians(90.0f), glm::vec3(1.0f, 0.0f, 0.0f));
398 model = glm::translate(model, glm::vec3(-1.3f, -3.65f, 2.6f));
399 model = glm::scale(model, glm::vec3(1.5f, 1.5f, 1.5f));
400 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion6()), glm::vec3(0.0f, 1.0f, 0.0f));
401 glm::mat4 modelRin2 = model;
402 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
403 color = glm::vec3(0.0f, 0.0f, 0.0f);
404 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
405 meshList[5]->RenderMeshGeometry();
406
407 //Centro de llanta
408 model = modelRin2;
409 model = glm::scale(model, glm::vec3(0.6f, 1.1f, 0.6f));
410 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
411 color = glm::vec3(1.0f, 1.0f, 1.0f); // Blanco
412 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
413 meshList[5]->RenderMeshGeometry();
414
415 //Llanta 3
416 model = modelaux;
417 model = glm::rotate(model, glm::radians(90.0f), glm::vec3(0.0f, 0.0f, 1.0f));
418 model = glm::rotate(model, glm::radians(90.0f), glm::vec3(1.0f, 0.0f, 0.0f));
419 model = glm::translate(model, glm::vec3(-1.3f, -3.65f, -2.6f));
420 model = glm::scale(model, glm::vec3(1.5f, 1.5f, 1.5f));
421 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion7()), glm::vec3(0.0f, 1.0f, 0.0f));
422 glm::mat4 modelRin3 = model;
423 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
424 color = glm::vec3(0.0f, 0.0f, 0.0f);

```

```

424     color = glm::vec3(0.0f, 0.0f, 0.0f);
425     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
426     meshList[5]->RenderMeshGeometry();
427
428     //Centro de llanta
429     model = modelRin3;
430     model = glm::scale(model, glm::vec3(0.6f, 1.1f, 0.6f));
431     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
432     color = glm::vec3(1.0f, 1.0f, 1.0f);
433     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
434     meshList[5]->RenderMeshGeometry();
435
436     //Llanta 4
437     model = modelaux;
438     model = glm::rotate(model, glm::radians(90.0f), glm::vec3(0.0f, 0.0f, 1.0f));
439     model = glm::rotate(model, glm::radians(90.0f), glm::vec3(1.0f, 0.0f, 0.0f));
440     model = glm::translate(model, glm::vec3(-1.3f, 3.65f, -2.6f));
441     model = glm::scale(model, glm::vec3(1.5f, 1.5f, 1.5f));
442     model = glm::rotate(model, glm::radians(mainWindow.getarticulacion8()), glm::vec3(0.0f, 1.0f, 0.0f));
443     glm::mat4 modelRin4 = model;
444     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
445     color = glm::vec3(0.0f, 0.0f, 0.0f);
446     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
447     meshList[5]->RenderMeshGeometry();
448
449     //Centro de llanta
450     model = modelRin4;
451     model = glm::scale(model, glm::vec3(0.6f, 1.1f, 0.6f));
452     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
453     color = glm::vec3(1.0f, 1.0f, 1.0f);
454     glUniform3fv(uniformColor, 1, glm::value_ptr(color));

```

```

453     color = glm::vec3(1.0f, 1.0f, 1.0f);
454     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
455     meshList[5]->RenderMeshGeometry();
456
457     //articulación 1
458     model = modelaux;
459     model = glm::rotate(model, glm::radians(mainWindow.getarticulacion1()), glm::vec3(0.0f, 0.0f, 1.0f));
460     model = glm::rotate(model, glm::radians(135.0f), glm::vec3(0.0f, 0.0f, 1.0f));
461     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
462
463     //Primer brazo
464     model = glm::translate(model, glm::vec3(2.5f, 0.0f, 0.0f));
465     modelaux = model;
466     model = glm::scale(model, glm::vec3(5.0f, 1.0f, 1.0f));
467     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
468     color = glm::vec3(1.0f, 1.0f, 0.5f);
469     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
470     meshList[0]->RenderMesh();
471
472     //articulacion 2
473     model = modelaux;
474     model = glm::translate(model, glm::vec3(2.5f, 0.0f, 0.0f));
475     model = glm::rotate(model, glm::radians(mainWindow.getarticulacion2()), glm::vec3(0.0f, 0.0f, 1.0f));
476     modelaux = model;
477     model = glm::scale(model, glm::vec3(0.5f, 0.5f, 0.5f));
478     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
479     color = glm::vec3(0.0f, 0.0f, 0.0f);
480     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
481     sp.render();

```

```

481 sp.render();
482
483 //segundo brazo
484 model = modelaux;
485 model = glm::translate(model, glm::vec3(0.0f, -2.5f, 0.0f));
486 modelaux = model;
487 model = glm::scale(model, glm::vec3(1.0f, 5.0f, 1.0f));
488 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
489 color = glm::vec3(1.0f, 1.0f, 0.5f);
490 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
491 meshList[0]->RenderMesh();
492
493 model = modelaux;
494
495 //Articulación 3
496 model = glm::translate(model, glm::vec3(0.0f, -2.5f, 0.0f));
497 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion3()), glm::vec3(0.0f, 0.0f, 1.0f));
498 modelaux = model;
499 model = glm::scale(model, glm::vec3(0.5f, 0.5f, 0.5f));
500 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
501 color = glm::vec3(0.0f, 0.0f, 0.0f);
502 color = glm::vec3(0.0f, 0.0f, 0.0f);
503 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
504 sp.render();|
505
506 //Tercer Brazo-----
507 model = modelaux;
508 model = glm::translate(model, glm::vec3(0.0f, -2.5f, 0.0f));
509 modelaux = model;
510 model = glm::scale(model, glm::vec3(1.0f, 5.0f, 1.0f));
511 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
512
513 model = glm::scale(model, glm::vec3(1.0f, 5.0f, 1.0f));
514 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
515 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
516 color = glm::vec3(1.0f, 1.0f, 0.5f);
517 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
518 meshList[0]->RenderMesh();
519
520 //Articulación 4-----
521 model = modelaux;
522 model = glm::translate(model, glm::vec3(0.0f, -2.5f, 0.0f));
523 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion4()), glm::vec3(0.0f, 0.0f, -1.0f));
524 modelaux = model;
525
526 model = glm::scale(model, glm::vec3(0.5f, 0.5f, 0.5f));
527 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
528 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
529 color = glm::vec3(0.0f, 0.0f, 0.0f);
530 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
531 sp.render();
532
533 //Canasta-----
534 model = modelaux;
535 model = glm::translate(model, glm::vec3(0.0f, -1.25f, 0.0f));
536 model = glm::scale(model, glm::vec3(2.5f, 2.5f, 2.5f));
537 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
538 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
539 color = glm::vec3(0.5f, 0.5f, 0.5f);
540 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
541 meshList[0]->RenderMesh();
542
543 //Fin Grúa-----

```

Para este primer ejercicio se implementa la estructura jerárquica principalmente a través del uso de una matriz de respaldo llamada modelaux, la cual se inicializa con las transformaciones del "nodo padre" (como la cabina de la grúa o el pecho del animal) y se hereda a los "nodos hijos" (como brazos o llantas) para que estos sigan el movimiento global de la base. En el caso específico de la grúa, la jerarquía se observa en la construcción de los brazos, donde cada nuevo segmento se posiciona mediante un glm::translate y un glm::rotate relativo a la posición del brazo anterior, asegurando que si la articulación inicial rota, toda la cadena mecánica responda de

forma coherente. Así mismo, para el sistema de llantas, la jerarquía permite que el "rin" blanco herede exactamente la matriz de transformación y rotación de la "llanta" negra, garantizando que ambos componentes giren como una sola unidad.

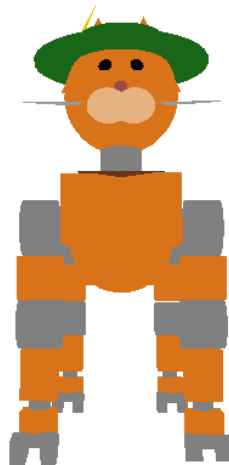
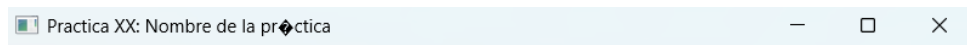
2.- Crear un animal robot 3d

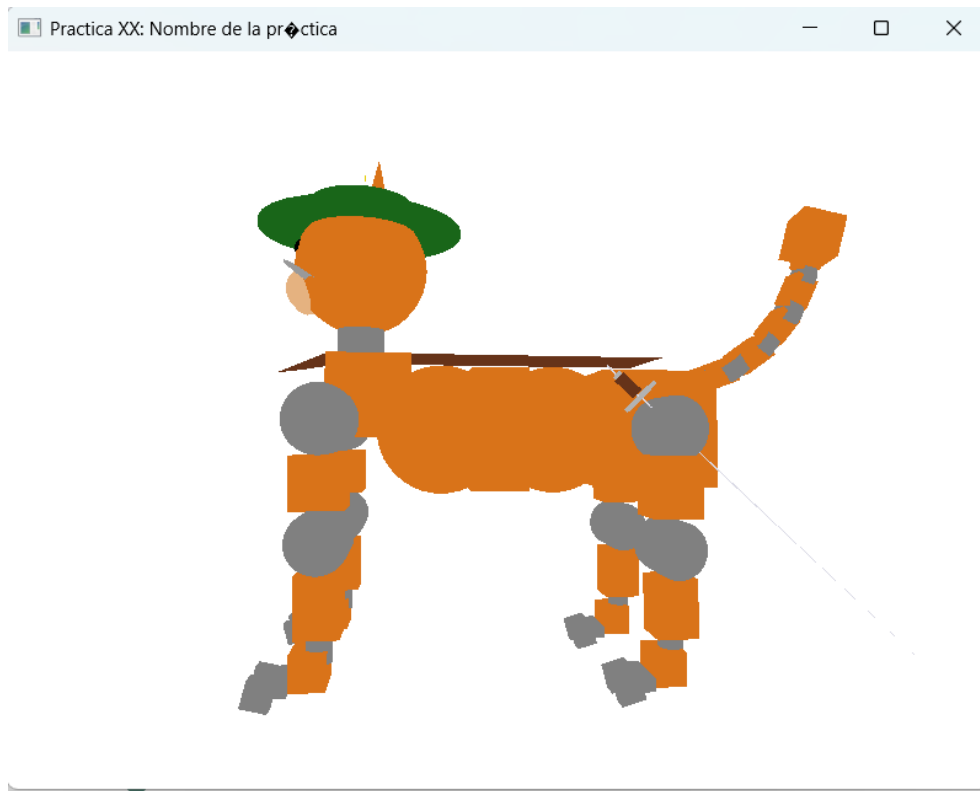
-Instanciando cubos, pirámides, cilindros, conos, esferas:

-4 patas articuladas en 2 partes (con teclado se puede mover las dos articulaciones de cada pata)

-cola articulada o 2 orejas articuladas. (con teclado se puede mover la cola o cada oreja independiente)

Capturas de ejecución:





Video de ejecución: <https://www.youtube.com/watch?v=PjmsRyquB4o>

Código de ejercicio:

```

543 //Torso-----
544 //Pecho
545 model = glm::mat4(1.0);
546 model = glm::translate(model, glm::vec3(0.0f, 7.0f, -4.0f));
547 //aquí está la jerarquía
548 modelaux = model;
549 model = glm::scale(model, glm::vec3(3.5f, 2.0f, 2.0f));
550 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
551 //la línea de proyección solo se manda una vez a menos que en tiempo de ejecución
552 //se programe cambio entre proyección ortogonal y perspectiva
553 glUniformMatrix4fv(uniformProjection, 1, GL_FALSE, glm::value_ptr(projection));
554 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
555 color = glm::vec3(0.85f, 0.45f, 0.1f); // Naranja
556 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
557 meshList[0]->RenderMesh();
558
559
560 //Primera esfera continuación de torso-----
561 model = modelaux;
562 model = glm::translate(model, glm::vec3(0.0f, -0.8f, -1.8f));
563 modelaux = model;
564 model = glm::scale(model, glm::vec3(1.7f, 1.7f, 1.7f));
565 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
566 sp.render();
567
568 //Cadera con rectangulo-----
569 model = modelaux;
570 model = glm::translate(model, glm::vec3(0.0f, 0.0f, -1.5f));
571 glm::mat4 modelauxCadera = model;
572 model = glm::scale(model, glm::vec3(1.5f, 3.2f, 1.5f));

```



```

572 |     model = glm::scale(model, glm::vec3(1.5f, 3.2f, 1.5f));
573 |     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
574 |     meshList[0]->RenderMesh();
575 |
576 |     //Segunda esfera continuación del torso-----
577 |     model = modelauxCadera;
578 |     model = glm::translate(model, glm::vec3(0.0f, 0.0f, -1.5f));
579 |     modelaux = model;
580 |     model = glm::scale(model, glm::vec3(1.7f, 1.7f, 1.7f));
581 |     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
582 |     sp.render();
583 |
584 |     //Cubo final del torso-----
585 |     model = modelaux;
586 |     model = glm::translate(model, glm::vec3(0.0f, 0.0f, -2.5f));
587 |     glm::mat4 modelauxFinal = model;
588 |     model = glm::scale(model, glm::vec3(3.0f, 3.0f, 3.0f));
589 |     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
590 |     meshList[0]->RenderMesh();
591 |
592 |     //Capa trasera-----
593 |     model = modelauxFinal;
594 |     model = glm::translate(model, glm::vec3(0.0f, 1.9f, 4.1f));
595 |     model = glm::rotate(model, glm::radians(180.0f), glm::vec3(0.0f, 1.0f, 0.0f));
596 |     color = glm::vec3(0.4f, 0.2f, 0.1f);
597 |     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
598 |     model = glm::scale(model, glm::vec3(2.5f, 0.1f, 8.5f));
599 |     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
600 |     meshList[4]->RenderMeshGeometry();
601 |     model = modelaux;

```

```

601 |     model = modelaux;
602 |     model = glm::translate(model, glm::vec3(0.0f, 2.2f, 5.1f));
603 |     model = glm::rotate(model, glm::radians(mainWindow.getarticulacion1()), glm::vec3(0.0f, 1.0f, 0.0f));
604 |     glm::mat4 modelauxCuello = model;
605 |     color = glm::vec3(0.5f, 0.5f, 0.5f);
606 |     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
607 |     model = glm::scale(model, glm::vec3(0.6f, 0.8f, 0.6f));
608 |     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
609 |     meshList[2]->RenderMeshGeometry();
610 |
611 |     //Capa delantera-----
612 |     model = modelauxCuello;
613 |     model = glm::translate(model, glm::vec3(0.0f, -0.4f, 0.7f));
614 |     model = glm::rotate(model, glm::radians(180.0f), glm::vec3(1.0f, 0.0f, 0.0f));
615 |     model = glm::rotate(model, glm::radians(15.0f), glm::vec3(1.0f, 0.0f, 0.0f));
616 |     color = glm::vec3(0.4f, 0.2f, 0.1f);
617 |     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
618 |     model = glm::scale(model, glm::vec3(2.5f, 0.1f, 3.0f));
619 |     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
620 |     meshList[1]->RenderMeshGeometry();
621 |
622 |     //Espada-----
623 |     model = modelauxCadera;
624 |     model = glm::translate(model, glm::vec3(2.2f, 0.5f, -3.5f));
625 |     model = glm::rotate(model, glm::radians(45.0f), glm::vec3(1.0f, 0.0f, 0.0f));
626 |     model = glm::rotate(model, glm::radians(10.0f), glm::vec3(0.0f, 0.0f, 1.0f));
627 |     glm::mat4 modelauxEspada = model;
628 |
629 |     //Mango de espada-----
630 |     color = glm::vec3(0.4f, 0.2f, 0.1f);

```

```

629 //Mango de espada-----
630 color = glm::vec3(0.4f, 0.2f, 0.1f);
631 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
632 model = modelauxEspada;
633 model = glm::translate(model, glm::vec3(0.0f, 0.8f, 0.0f));
634 model = glm::scale(model, glm::vec3(0.15f, 0.6f, 0.15f));
635 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
636 meshList[2]->RenderMeshGeometry();
637
638 //Guarda de espada-----
639 color = glm::vec3(0.7f, 0.7f, 0.7f);
640 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
641 model = modelauxEspada;
642 model = glm::translate(model, glm::vec3(0.0f, 0.4f, 0.0f));
643 model = glm::scale(model, glm::vec3(0.5f, 0.1f, 0.5f));
644 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
645 meshList[2]->RenderMeshGeometry();
646
647 //Hoja de espada-----
648 color = glm::vec3(0.85f, 0.85f, 0.9f);
649 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
650 model = modelauxEspada;
651 model = glm::translate(model, glm::vec3(0.0f, -3.5f, 0.0f));
652 model = glm::rotate(model, glm::radians(180.0f), glm::vec3(1.0f, 0.0f, 0.0f));
653 model = glm::scale(model, glm::vec3(0.15f, 10.0f, 0.05f));
654 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
655 meshList[1]->RenderMeshGeometry();
656
657 //Pomo de espada-----
658 color = glm::vec3(0.7f, 0.7f, 0.7f);

```

```

657 //Pomo de espada-----
658 color = glm::vec3(0.7f, 0.7f, 0.7f);
659 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
660 model = modelauxEspada;
661 model = glm::translate(model, glm::vec3(0.0f, 1.1f, 0.0f));
662 model = glm::scale(model, glm::vec3(0.2f, 0.2f, 0.2f));
663 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
664 meshList[0]->RenderMesh();
665
666 //Cabeza-----
667 model = modelauxCuello;
668 model = glm::translate(model, glm::vec3(0.0f, 2.0f, 0.0f));
669 glm::mat4 modelauxCabeza = model;
670
671 color = glm::vec3(0.85f, 0.45f, 0.1f);
672 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
673 model = glm::scale(model, glm::vec3(1.7f, 1.7f, 1.7f));
674 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
675 sp.render(); // Render de la esfera base
676
677 //Ojos de cabeza-----
678 color = glm::vec3(0.0f, 0.0f, 0.0f);
679 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
680
681 //Ojo Izquierdo-----
682 model = modelauxCabeza;
683 model = glm::translate(model, glm::vec3(0.4f, 0.4f, 1.4f));
684 model = glm::scale(model, glm::vec3(0.25f, 0.25f, 0.25f));
685 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
686 sp.render(); //
687

```

```

687
688 //Ojo Derecho-----
689 model = modelauxCabeza;
690 model = glm::translate(model, glm::vec3(-0.4f, 0.4f, 1.4f));
691 model = glm::scale(model, glm::vec3(0.25f, 0.25f, 0.25f));
692 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
693 sp.render();
694
695 //Orejas-----
696 color = glm::vec3(0.85f, 0.45f, 0.1f);
697 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
698
699 //Oreja Izquierda-----
700 model = modelauxCabeza;
701 model = glm::translate(model, glm::vec3(0.7f, 1.4f, 0.0f));
702 model = glm::scale(model, glm::vec3(2.2f, 2.5f, 1.9f));
703 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
704 meshList[1]->RenderMeshGeometry();
705
706 //Oreja Derecha-----
707 model = modelauxCabeza;
708 model = glm::translate(model, glm::vec3(-0.7f, 1.4f, 0.0f));
709 model = glm::scale(model, glm::vec3(2.2f, 2.5f, 1.9f));
710 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
711 meshList[1]->RenderMeshGeometry();
712
713 //Sombrero-----
714 color = glm::vec3(0.1f, 0.4f, 0.1f);
715 glUniform3fv(uniformColor, 1, glm::value_ptr(color));

```

```

715 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
716
717 //Ala del sombrero-----
718 model = modelauxCabeza;
719 model = glm::translate(model, glm::vec3(0.0f, 1.0f, 0.0f));
720 glm::mat4 modelauxAla = model;
721 model = glm::scale(model, glm::vec3(2.6f, 0.1f, 2.6f));
722 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
723 meshList[2]->RenderMeshGeometry();
724
725 //Copa del sombrero-----
726 model = modelauxAla;
727 model = glm::translate(model, glm::vec3(0.0f, 0.4f, 0.0f));
728 model = glm::scale(model, glm::vec3(1.3f, 0.7f, 1.3f));
729 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
730 meshList[2]->RenderMeshGeometry();
731
732 //Pluma del sombrero-----
733 color = glm::vec3(0.9f, 0.8f, 0.0f);
734 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
735 model = modelauxAla;
736 model = glm::translate(model, glm::vec3(-1.0f, 1.3f, 0.0f));
737 model = glm::rotate(model, glm::radians(-25.0f), glm::vec3(0.0f, 0.0f, 1.0f));
738 model = glm::scale(model, glm::vec3(0.15f, 1.4f, 0.05f));
739 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
740 meshList[1]->RenderMeshGeometry();
741
742 //Bigotes-----
743 color = glm::vec3(0.6f, 0.6f, 0.6f);
744 glUniform3fv(uniformColor, 1, glm::value_ptr(color));

```

```

744 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
745
746 //Bigote izquierdo-----
747 model = modelauxCabeza;
748 model = glm::translate(model, glm::vec3(0.8f, -0.5f, 1.1f));
749 model = glm::rotate(model, glm::radians(80.0f), glm::vec3(0.0f, 1.0f, 0.0f));
750 model = glm::scale(model, glm::vec3(0.08f, 0.08f, 1.0f));
751 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
752 meshList[3]->RenderMeshGeometry(); // Cono
753
754 //Bigote derecho-----
755 model = modelauxCabeza;
756 model = glm::translate(model, glm::vec3(-0.8f, -0.5f, 1.1f));
757 model = glm::rotate(model, glm::radians(-80.0f), glm::vec3(0.0f, 1.0f, 0.0f));
758 model = glm::scale(model, glm::vec3(0.08f, 0.08f, 1.0f));
759 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
760 meshList[3]->RenderMeshGeometry(); // Cono
761
762 //Boca-----
763 color = glm::vec3(0.9f, 0.7f, 0.5f);
764 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
765
766 //Parte izquierda de la boca-----
767 model = modelauxCabeza;
768 model = glm::translate(model, glm::vec3(0.35f, -0.65f, 1.45f));
769 model = glm::scale(model, glm::vec3(0.6f, 0.5f, 0.4f));
770 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
771 sp.render();
772
773 //Parte derecha de la boca-----
774 model = modelauxCabeza;

```

```

774 model = modelauxCabeza;
775 model = glm::translate(model, glm::vec3(-0.35f, -0.65f, 1.45f));
776 model = glm::scale(model, glm::vec3(0.6f, 0.5f, 0.4f));
777 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
778 sp.render();
779
780 //Nariz-----
781 model = modelauxCabeza;
782 model = glm::translate(model, glm::vec3(0.0f, -0.15f, 1.6f));
783 model = glm::rotate(model, glm::radians(90.0f), glm::vec3(1.0f, 0.0f, 0.0f));
784 color = glm::vec3(0.6f, 0.3f, 0.3f);
785 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
786 model = glm::scale(model, glm::vec3(0.3f, 0.3f, 0.3f));
787 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
788 meshList[3]->RenderMeshGeometry();
789
790 //Pata delantera izquierda-----
791
792 model = glm::mat4(1.0);
793 model = glm::translate(model, glm::vec3(1.1f, 7.4f, -4.0f));
794 model = glm::translate(model, glm::vec3(1.2f, -1.0f, 1.0f));
795 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion2()), glm::vec3(1.0f, 0.0f, 0.0f));
796 glm::mat4 modelauxArt1 = model;
797 color = glm::vec3(0.5f, 0.5f, 0.5f);
798 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
799 model = glm::rotate(model, glm::radians(90.0f), glm::vec3(0.0f, 0.0f, 1.0f));
800 model = glm::scale(model, glm::vec3(0.8f, 1.0f, 0.8));
801 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
802 meshList[2]->RenderMeshGeometry();
803 model = modelauxArt1;

```

```

805     glm::mat4 modelauxBrazoSup = model;
806     color = glm::vec3(0.85f, 0.45f, 0.1f);
807     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
808     model = glm::scale(model, glm::vec3(1.2f, 1.2f, 1.2f));
809     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
810     meshList[0]->RenderMesh();
811     model = modelauxBrazoSup;
812     model = glm::translate(model, glm::vec3(0.0f, -1.3f, 0.0f));
813     model = glm::rotate(model, glm::radians(mainWindow.getarticulacion3()), glm::vec3(1.0f, 0.0f, 0.0f));
814     glm::mat4 modelauxArt2 = model;
815     color = glm::vec3(0.5f, 0.5f, 0.5f);
816     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
817     model = glm::rotate(model, glm::radians(90.0f), glm::vec3(0.0f, 0.0f, 1.0f));
818     model = glm::scale(model, glm::vec3(0.7f, 1.2f, 0.7f));
819     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
820     meshList[2]->RenderMeshGeometry();
821     model = modelauxArt2;
822     model = glm::translate(model, glm::vec3(0.0f, -1.4f, 0.0f));
823     glm::mat4 modelauxBrazoInf = model;
824     color = glm::vec3(0.85f, 0.45f, 0.1f);
825     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
826     model = glm::scale(model, glm::vec3(1.0f, 1.4f, 1.0f));
827     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
828     meshList[0]->RenderMesh();
829     model = modelauxBrazoInf;
830     model = glm::translate(model, glm::vec3(0.0f, -1.0f, 0.0f));
831     model = glm::rotate(model, glm::radians(mainWindow.getarticulacion4()), glm::vec3(1.0f, 0.0f, 0.0f));
832     glm::mat4 modelauxArt3 = model;
833     color = glm::vec3(0.5f, 0.5f, 0.5f);
834     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
835     model = glm::scale(model, glm::vec3(0.3f, 0.6f, 0.3f));

```

```

835     model = glm::scale(model, glm::vec3(0.3f, 0.6f, 0.3f));
836     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
837     meshList[2]->RenderMeshGeometry();
838
839     //Pie-----
840     model = modelauxArt3;
841     model = glm::translate(model, glm::vec3(0.0f, -0.4f, 0.2f));
842     glm::mat4 modelauxMaderaPie = model;
843     color = glm::vec3(0.85f, 0.45f, 0.1f);
844     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
845     model = glm::scale(model, glm::vec3(1.0f, 0.8f, 0.8f));
846     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
847     meshList[0]->RenderMesh();
848     model = modelauxMaderaPie;
849     model = glm::translate(model, glm::vec3(0.0f, -0.3f, 0.6f));
850     glm::mat4 modelauxSoporteMetal = model;
851     color = glm::vec3(0.5f, 0.5f, 0.5f);
852     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
853     model = glm::scale(model, glm::vec3(1.2f, 0.3f, 0.4f));
854     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
855     meshList[0]->RenderMesh();
856     model = modelauxSoporteMetal;
857     model = glm::translate(model, glm::vec3(-0.4f, -0.1f, 0.4f));
858     model = glm::rotate(model, glm::radians(-15.0f), glm::vec3(1.0f, 0.0f, 0.0f));
859     color = glm::vec3(0.5f, 0.5f, 0.5f);
860     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
861     model = glm::scale(model, glm::vec3(0.4f, 0.6f, 0.6f));
862     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
863     meshList[0]->RenderMesh();
864     model = modelauxSoporteMetal;

```

```

864 model = modelauxSoporteMetal;
865 model = glm::translate(model, glm::vec3(0.4f, -0.1f, 0.4f));
866 model = glm::rotate(model, glm::radians(-15.0f), glm::vec3(1.0f, 0.0f, 0.0f));
867 color = glm::vec3(0.5f, 0.5f, 0.5f);
868 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
869 model = glm::scale(model, glm::vec3(0.4f, 0.6f, 0.6f));
870 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
871 meshList[0]->RenderMesh();
872
873 //Pata delantera derecha-----
874 model = glm::mat4(1.0);
875 model = glm::translate(model, glm::vec3(-1.1f, 7.4f, -4.0f));
876 model = glm::translate(model, glm::vec3(-1.2f, -1.0f, 1.0f));
877 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion5()), glm::vec3(1.0f, 0.0f, 0.0f));
878 glm::mat4 modelauxArt1Der = model;
879 color = glm::vec3(0.5f, 0.5f, 0.5f);
880 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
881 model = glm::rotate(model, glm::radians(90.0f), glm::vec3(0.0f, 0.0f, 1.0f));
882 model = glm::scale(model, glm::vec3(0.8f, 1.0f, 0.8f));
883 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
884 meshList[2]->RenderMeshGeometry();
885 model = modelauxArt1Der;
886 model = glm::translate(model, glm::vec3(0.0f, -1.4f, 0.0f));
887 glm::mat4 modelauxBrazoSupDer = model;
888 color = glm::vec3(0.85f, 0.45f, 0.1f);
889 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
890 model = glm::scale(model, glm::vec3(1.2f, 1.2f, 1.2f));
891 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
892 meshList[0]->RenderMesh();
893 model = modelauxBrazoSupDer;
894 model = glm::translate(model, glm::vec3(0.0f, -1.3f, 0.0f));

```

```

894 model = glm::translate(model, glm::vec3(0.0f, -1.3f, 0.0f));
895 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion6()), glm::vec3(1.0f, 0.0f, 0.0f));
896 glm::mat4 modelauxArt2Der = model;
897 color = glm::vec3(0.5f, 0.5f, 0.5f);
898 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
899 model = glm::rotate(model, glm::radians(90.0f), glm::vec3(0.0f, 0.0f, 1.0f));
900 model = glm::scale(model, glm::vec3(0.7f, 1.2f, 0.7f));
901 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
902 meshList[2]->RenderMeshGeometry();
903 model = modelauxArt2Der;
904 model = glm::translate(model, glm::vec3(0.0f, -1.4f, 0.0f));
905 glm::mat4 modelauxBrazoInfDer = model;
906 color = glm::vec3(0.85f, 0.45f, 0.1f);
907 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
908 model = glm::scale(model, glm::vec3(1.0f, 1.4f, 1.0f));
909 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
910 meshList[0]->RenderMesh();
911 model = modelauxBrazoInfDer;
912 model = glm::translate(model, glm::vec3(0.0f, -1.0f, 0.0f));
913 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion7()), glm::vec3(1.0f, 0.0f, 0.0f));
914 glm::mat4 modelauxArt3Der = model;
915 color = glm::vec3(0.5f, 0.5f, 0.5f);
916 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
917 model = glm::scale(model, glm::vec3(0.3f, 0.6f, 0.3f));
918 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
919 meshList[2]->RenderMeshGeometry();
920
921 //Pie-----
922 model = modelauxArt3Der;
923 model = glm::translate(model, glm::vec3(0.0f, -0.4f, 0.2f));

```



```

923     model = glm::translate(model, glm::vec3(0.0f, -0.4f, 0.2f));
924     glm::mat4 modelauxMaderaPieDer = model;
925     color = glm::vec3(0.85f, 0.45f, 0.1f);
926     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
927     model = glm::scale(model, glm::vec3(1.0f, 0.8f, 0.8f));
928     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
929     meshList[0]->RenderMesh();
930     model = modelauxMaderaPieDer;
931     model = glm::translate(model, glm::vec3(0.0f, -0.3f, 0.6f));
932     glm::mat4 modelauxSoporteMetalDer = model;
933     color = glm::vec3(0.5f, 0.5f, 0.5f);
934     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
935     model = glm::scale(model, glm::vec3(1.2f, 0.3f, 0.4f));
936     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
937     meshList[0]->RenderMesh();
938     model = modelauxSoporteMetalDer;
939     model = glm::translate(model, glm::vec3(0.4f, -0.1f, 0.4f));
940     model = glm::rotate(model, glm::radians(15.0f), glm::vec3(1.0f, 0.0f, 0.0f));
941     color = glm::vec3(0.5f, 0.5f, 0.5f);
942     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
943     model = glm::scale(model, glm::vec3(0.4f, 0.6f, 0.6f));
944     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
945     meshList[0]->RenderMesh();
946     model = modelauxSoporteMetalDer;
947     model = glm::translate(model, glm::vec3(-0.4f, -0.1f, 0.4f));
948     model = glm::rotate(model, glm::radians(15.0f), glm::vec3(1.0f, 0.0f, 0.0f));
949     color = glm::vec3(0.5f, 0.5f, 0.5f);
950     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
951     model = glm::scale(model, glm::vec3(0.4f, 0.6f, 0.6f));
952     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
953     meshList[0]->RenderMesh();

```

```

953     meshList[0]->RenderMesh();
954
955     //Pata trasera izquierda-----
956     model = modelauxCadera;
957     model = glm::translate(model, glm::vec3(2.0f, 0.0f, -4.0f));
958     model = glm::rotate(model, glm::radians(mainWindow.getarticulacion8()), glm::vec3(1.0f, 0.0f, 0.0f));
959     glm::mat4 modelauxArtTras1 = model;
960     color = glm::vec3(0.5f, 0.5f, 0.5f);
961     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
962     model = glm::rotate(model, glm::radians(90.0f), glm::vec3(0.0f, 0.0f, 1.0f));
963     model = glm::scale(model, glm::vec3(0.8f, 1.0f, 0.8));
964     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
965     meshList[2]->RenderMeshGeometry();
966     model = modelauxArtTras1;
967     model = glm::translate(model, glm::vec3(0.0f, -1.4f, 0.0f));
968     glm::mat4 modelauxMuslo = model;
969     color = glm::vec3(0.85f, 0.45f, 0.1f);
970     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
971     model = glm::scale(model, glm::vec3(1.2f, 1.5f, 1.2f));
972     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
973     meshList[0]->RenderMesh();
974     model = modelauxMuslo;
975     model = glm::translate(model, glm::vec3(0.0f, -1.5f, 0.0f));
976     model = glm::rotate(model, glm::radians(mainWindow.getarticulacion9()), glm::vec3(1.0f, 0.0f, 0.0f));
977     glm::mat4 modelauxArtTras2 = model;
978     color = glm::vec3(0.5f, 0.5f, 0.5f);
979     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
980     model = glm::rotate(model, glm::radians(90.0f), glm::vec3(0.0f, 0.0f, 1.0f));
981     model = glm::scale(model, glm::vec3(0.7f, 1.2f, 0.7f));
982     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
983     meshList[2]->RenderMeshGeometry();

```

```

983 meshList[2]->RenderMeshGeometry();
984 model = modelauxArtTras2;
985 model = glm::translate(model, glm::vec3(0.0f, -1.4f, 0.0f));
986 glm::mat4 modelauxPiernaTras = model;
987 color = glm::vec3(0.85f, 0.45f, 0.1f);
988 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
989 model = glm::scale(model, glm::vec3(1.0f, 1.4f, 1.0f));
990 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
991 meshList[0]->RenderMesh();
992 model = modelauxPiernaTras;
993 model = glm::translate(model, glm::vec3(0.0f, -1.0f, 0.0f));
994 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion10()), glm::vec3(1.0f, 0.0f, 0.0f));
995 glm::mat4 modelauxArtTras3 = model;
996 color = glm::vec3(0.5f, 0.5f, 0.5f);
997 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
998 model = glm::scale(model, glm::vec3(0.3f, 0.6f, 0.3f));
999 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1000 meshList[2]->RenderMeshGeometry();
1001
1002 //Pie
1003 model = modelauxArtTras3;
1004 model = glm::translate(model, glm::vec3(0.0f, -0.4f, 0.2f));
1005 glm::mat4 modelauxMaderaPieTras = model;
1006 color = glm::vec3(0.85f, 0.45f, 0.1f);
1007 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1008 model = glm::scale(model, glm::vec3(0.8f));
1009 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1010 meshList[0]->RenderMesh();
1011 model = modelauxMaderaPieTras;
1012 model = glm::translate(model, glm::vec3(0.0f, -0.3f, 0.6f));
1013 glm::mat4 modelauxSoporteMetalTras = model;

```

```

1012 model = glm::translate(model, glm::vec3(0.0f, -0.3f, 0.6f));
1013 glm::mat4 modelauxSoporteMetalTras = model;
1014 color = glm::vec3(0.5f, 0.5f, 0.5f);
1015 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1016 model = glm::scale(model, glm::vec3(1.2f, 0.3f, 0.4f));
1017 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1018 meshList[0]->RenderMesh();
1019 model = modelauxSoporteMetalTras;
1020 model = glm::translate(model, glm::vec3(0.4f, -0.1f, 0.4f));
1021 model = glm::rotate(model, glm::radians(15.0f), glm::vec3(1.0f, 0.0f, 0.0f));
1022 model = glm::scale(model, glm::vec3(0.4f, 0.6f, 0.6f));
1023 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1024 meshList[0]->RenderMesh();
1025 model = modelauxSoporteMetalTras;
1026 model = glm::translate(model, glm::vec3(-0.4f, -0.1f, 0.4f));
1027 model = glm::rotate(model, glm::radians(15.0f), glm::vec3(1.0f, 0.0f, 0.0f));
1028 model = glm::scale(model, glm::vec3(0.4f, 0.6f, 0.6f));
1029 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1030 meshList[0]->RenderMesh(); // Cubo
1031
1032 //Pata trasera derecha-----
1033 model = modelauxCadera;
1034 model = glm::translate(model, glm::vec3(-2.1f, 0.0f, -4.0f));
1035 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion11()), glm::vec3(1.0f, 0.0f, 0.0f));
1036 glm::mat4 modelauxArtTras1Der = model;
1037 color = glm::vec3(0.5f, 0.5f, 0.5f);
1038 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1039 model = glm::rotate(model, glm::radians(90.0f), glm::vec3(0.0f, 0.0f, 1.0f));
1040 model = glm::scale(model, glm::vec3(0.8f, 1.0f, 0.8));
1041 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1042 meshList[2]->RenderMeshGeometry();

```



```

1041 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1042 meshList[2]->RenderMeshGeometry();
1043 model = modelauxArtTras1Der;
1044 model = glm::translate(model, glm::vec3(0.0f, -1.4f, 0.0f));
1045 glm::mat4 modelauxMusloDer = model;
1046 color = glm::vec3(0.85f, 0.45f, 0.1f);
1047 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1048 model = glm::scale(model, glm::vec3(1.2f, 1.5f, 1.2f));
1049 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1050 meshList[0]->RenderMesh();
1051 model = modelauxMusloDer;
1052 model = glm::translate(model, glm::vec3(0.0f, -1.5f, 0.0f));
1053 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion12()), glm::vec3(1.0f, 0.0f, 0.0f));
1054 glm::mat4 modelauxArtTras2Der = model;
1055 color = glm::vec3(0.5f, 0.5f, 0.5f);
1056 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1057 model = glm::rotate(model, glm::radians(90.0f), glm::vec3(0.0f, 0.0f, 1.0f));
1058 model = glm::scale(model, glm::vec3(0.7f, 1.2f, 0.7f));
1059 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1060 meshList[2]->RenderMeshGeometry();
1061 model = modelauxArtTras2Der;
1062 model = glm::translate(model, glm::vec3(0.0f, -1.4f, 0.0f));
1063 glm::mat4 modelauxPiernaTrasDer = model;
1064 color = glm::vec3(0.85f, 0.45f, 0.1f);
1065 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1066 model = glm::scale(model, glm::vec3(1.0f, 1.4f, 1.0f));
1067 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1068 meshList[0]->RenderMesh();
1069 model = modelauxPiernaTrasDer;
1070 model = glm::translate(model, glm::vec3(0.0f, -1.0f, 0.0f));
1071 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion13()), glm::vec3(1.0f, 0.0f, 0.0f));

```

```

1071 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion13()), glm::vec3(1.0f, 0.0f, 0.0f));
1072 glm::mat4 modelauxArtTras3Der = model;
1073 color = glm::vec3(0.5f, 0.5f, 0.5f);
1074 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1075 model = glm::scale(model, glm::vec3(0.3f, 0.6f, 0.3f));
1076 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1077 meshList[2]->RenderMeshGeometry();
1078
1079 //Pie
1080 model = modelauxArtTras3Der;
1081 model = glm::translate(model, glm::vec3(0.0f, -0.4f, 0.2f));
1082 glm::mat4 modelauxMaderaPieTrasDer = model;
1083 color = glm::vec3(0.85f, 0.45f, 0.1f);
1084 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1085 model = glm::scale(model, glm::vec3(1.0f, 0.8f, 0.8f));
1086 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1087 meshList[0]->RenderMesh();
1088 model = modelauxMaderaPieTrasDer;
1089 model = glm::translate(model, glm::vec3(0.0f, -0.3f, 0.6f));
1090 glm::mat4 modelauxSoporteMetalTrasDer = model;
1091 color = glm::vec3(0.5f, 0.5f, 0.5f);
1092 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1093 model = glm::scale(model, glm::vec3(1.2f, 0.3f, 0.4f));
1094 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1095 meshList[0]->RenderMesh();
1096 model = modelauxSoporteMetalTrasDer;
1097 model = glm::translate(model, glm::vec3(0.4f, -0.1f, 0.4f));
1098 model = glm::rotate(model, glm::radians(15.0f), glm::vec3(1.0f, 0.0f, 0.0f));
1099 model = glm::scale(model, glm::vec3(0.4f, 0.6f, 0.6f));
1100 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1101 meshList[0]->RenderMesh();

```

```

1101 meshList[0]->RenderMesh();
1102 model = modelauxSoporteMetalTrasDer;
1103 model = glm::translate(model, glm::vec3(-0.4f, -0.1f, 0.4f));
1104 model = glm::rotate(model, glm::radians(15.0f), glm::vec3(1.0f, 0.0f, 0.0f));
1105 model = glm::scale(model, glm::vec3(0.4f, 0.6f, 0.6f));
1106 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
1107 meshList[0]->RenderMesh();
1108
1109 //Cola-----
1110 glm::mat4 modelTemp;
1111 model = modelauxFinal;
1112 model = glm::translate(model, glm::vec3(0.0f, 1.3f, -1.5f));
1113
1114 //Parte 1 de la cola-----
1115 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion14() + 20.0f), glm::vec3(1.0f, 0.0f, 0.0f));
1116
1117 color = glm::vec3(0.5f, 0.5f, 0.5f);
1118 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1119 modelTemp = glm::scale(model, glm::vec3(0.5f, 0.5f, 0.5f));
1120 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(modelTemp));
1121 meshList[2]->RenderMeshGeometry();
1122 model = glm::translate(model, glm::vec3(0.0f, 0.0f, -0.7f));
1123 color = glm::vec3(0.85f, 0.45f, 0.1f);
1124 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1125 modelTemp = glm::scale(model, glm::vec3(0.7f, 0.7f, 1.0f));
1126 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(modelTemp));
1127 meshList[0]->RenderMesh();
1128
1129 //Parte 2 de la cola-----
1130 model = glm::translate(model, glm::vec3(0.0f, 0.0f, -0.5f));
1131 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion15() + 15.0f), glm::vec3(1.0f, 0.0f, 0.0f));

```

```

1131 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion15() + 15.0f), glm::vec3(1.0f, 0.0f, 0.0f));
1132 color = glm::vec3(0.5f, 0.5f, 0.5f);
1133 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1134 modelTemp = glm::scale(model, glm::vec3(0.5f, 0.5f, 0.5f));
1135 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(modelTemp));
1136 meshList[2]->RenderMeshGeometry();
1137 model = glm::translate(model, glm::vec3(0.0f, 0.0f, -0.7f));
1138 color = glm::vec3(0.85f, 0.45f, 0.1f);
1139 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1140 modelTemp = glm::scale(model, glm::vec3(0.7f, 0.7f, 1.0f));
1141 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(modelTemp));
1142 meshList[0]->RenderMesh();
1143
1144 //Parte 3 de la cola-----
1145 model = glm::translate(model, glm::vec3(0.0f, 0.0f, -0.5f));
1146 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion16() + 15.0f), glm::vec3(1.0f, 0.0f, 0.0f));
1147 color = glm::vec3(0.5f, 0.5f, 0.5f);
1148 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1149 modelTemp = glm::scale(model, glm::vec3(0.45f, 0.45f, 0.45f));
1150 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(modelTemp));
1151 meshList[2]->RenderMeshGeometry();
1152 model = glm::translate(model, glm::vec3(0.0f, 0.0f, -0.7f));
1153 color = glm::vec3(0.85f, 0.45f, 0.1f);
1154 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1155 modelTemp = glm::scale(model, glm::vec3(0.65f, 0.65f, 0.9f));
1156 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(modelTemp));
1157 meshList[0]->RenderMesh();
1158
1159 //Parte 4 de la cola-----
1160 model = glm::translate(model, glm::vec3(0.0f, 0.0f, -0.5f));
1161 model = glm::rotate(model, glm::radians(mainWindow.getarticulacion17() + 15.0f), glm::vec3(1.0f, 0.0f, 0.0f));

```

```

1161     model = glm::rotate(model, glm::radians(mainWindow.getarticulacion17() + 15.0f), glm::vec3(1.0f, 0.0f, 0.0f));
1162     color = glm::vec3(0.5f, 0.5f, 0.5f);
1163     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1164     modelTemp = glm::scale(model, glm::vec3(0.45f, 0.45f, 0.45f));
1165     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(modelTemp));
1166     meshList[2] -> RenderMeshGeometry();
1167     model = glm::translate(model, glm::vec3(0.0f, 0.0f, -0.7f));
1168     color = glm::vec3(0.85f, 0.45f, 0.1f);
1169     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1170     modelTemp = glm::scale(model, glm::vec3(0.65f, 0.65f, 0.9f));
1171     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(modelTemp));
1172     meshList[0] -> RenderMesh();
1173
1174     //Parte 5 de la cola-----
1175     model = glm::translate(model, glm::vec3(0.0f, 0.0f, -0.5f));
1176     model = glm::rotate(model, glm::radians(mainWindow.getarticulacion18() + 10.0f), glm::vec3(1.0f, 0.0f, 0.0f));
1177     color = glm::vec3(0.5f, 0.5f, 0.5f);
1178     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1179     modelTemp = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
1180     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(modelTemp));
1181     meshList[2] -> RenderMeshGeometry();
1182     model = glm::translate(model, glm::vec3(0.0f, 0.0f, -0.7f));
1183     color = glm::vec3(0.85f, 0.45f, 0.1f);
1184     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1185     modelTemp = glm::scale(model, glm::vec3(0.6f, 0.6f, 0.9f));
1186     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(modelTemp));
1187     meshList[0] -> RenderMesh();
1188
1189     //Parte 6 de la cola-----
1190     model = glm::translate(model, glm::vec3(0.0f, 0.0f, -0.5f));
1191     model = glm::rotate(model, glm::radians(mainWindow.getarticulacion19() + 10.0f), glm::vec3(0.0f, 0.0f, 1.0f));
1192
1193     //Parte 6 de la cola-----
1194     model = glm::translate(model, glm::vec3(0.0f, 0.0f, -0.5f));
1195     model = glm::rotate(model, glm::radians(mainWindow.getarticulacion19() + 10.0f), glm::vec3(0.0f, 0.0f, 1.0f));
1196     color = glm::vec3(0.85f, 0.45f, 0.1f);
1197     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
1198     modelTemp = glm::scale(model, glm::vec3(1.2f, 1.2f, 1.2f)); // Punta más grande
1199     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(modelTemp));
1200     meshList[0] -> RenderMesh();
1201
1202     glUseProgram(0);
1203     mainWindow.swapBuffers();
1204 }
1205 return 0;
1206 }

```

Para este segundo ejercicio, en primer lugar, para la visualización previa del modelo jerárquico, se utilizaron herramientas de IA generativa con el fin de establecer una guía visual de la estructura geométrica.

Prompt 1: Definición de Estructura y Primitivas

"Crea una imagen de un animal robot 3D basado en el Gato con Botas de Dreamworks. El diseño debe estar construido exclusivamente mediante la instanciación de primitivas geométricas: cubos, pirámides, cilindros, conos y esferas. Debe mostrar claramente 4 patas articuladas en dos partes y una cola articulada, con indicadores visuales de control por teclado para cada articulación."

Prompt 2: Refinamiento de Estilo y Texturizado

"Genera una versión mejorada del gato robot anterior, manteniendo estrictamente el uso de cubos, cilindros y pirámides para su construcción. Aumenta el parecido visual con el personaje de Dreamworks (sombrero verde con pluma, capa, botas y espada) pero conservando la estética de un modelo de computación gráfica

jerárquico. Las articulaciones de las patas y la cola deben ser mecánicamente visibles para servir como guía de programación."

Imagen generada:



Figura 1. Concept art del Gato con Botas robótico. Generada por la IA Gemini (Google, 2026) a partir de instrucciones específicas sobre modelado jerárquico y primitivas geométricas.

Para este ejercicio, utilicé como referencia visual la imagen generada del Gato con Botas, implementando el animal robot 3D mediante una estructura de modelado jerárquico en OpenGL. En este caso, definí el torso como el nodo raíz (padre principal), lo que me permitió que todas las transformaciones posteriores dependieran de este punto central. Se gestiono la jerarquía a través de matrices auxiliares (modelaux) para asegurar que los movimientos se propaguen de forma coherente por lo que, de esta manera, el cuerpo queda vinculado con una cola articulada de seis niveles y extremidades de topología compleja.

El modelo cumple con la instanciación de primitivas solicitada, empleando cubos para el torso y las patas, esferas para las articulaciones de la cadera, y cilindros para el cuello y los segmentos de las extremidades. Así mismo, utilicé pirámides y conos para detallar las orejas, los bigotes y la pluma del sombrero. Diseñé las cuatro patas con una estructura jerárquica que incluye dos articulaciones móviles independientes, las cuales programé para ser controladas mediante el teclado. Finalmente, añadí los segmentos necesarios para que la cola tuviera articulaciones propias y un movimiento fluido.

2.- Liste los problemas que tuvo a la hora de hacer estos ejercicios y si los resolvió explicar cómo fue, en caso de error adjuntar captura de pantalla

Durante el desarrollo esta práctica, los principales problemas surgieron en el Ejercicio 1 al configurar el brazo de la grúa. Inicialmente, se presentaron dificultades en la implementación de la jerarquía de transformaciones, lo que impedía que los segmentos del brazo se movieran de manera dependiente y coherente. Fue necesario corregir el flujo de las matrices para asegurar que el movimiento de la base afectara correctamente a las otras partes del brazo.

Por otro lado, el Ejercicio 2 no presentó complicaciones estructurales mayores gracias a la experiencia previa con el modelo jerárquico. El mayor problema fue el proceso de experimentación y ajuste de coordenadas para cada uno de los componentes del Gato con Botas. Este proceso se realizó mediante un método de “ensayo y error”, calculando manualmente los desplazamientos y escalas necesarios para que las primitivas (cubos, cilindros y esferas) encajaran con precisión técnica y estética.

3.- Conclusión:

Tras la realización de esta práctica, considero que, más allá de la complejidad técnica, el desafío principal radicó en la inversión de tiempo requerida para cada ejercicio. El proceso de calcular y ajustar manualmente las coordenadas de cada elemento mediante experimentación es una tarea laboriosa; sin embargo, resultó ser un ejercicio fundamental para dominar el modelado jerárquico. Esta metodología me permitió aplicar y consolidar de manera práctica los conceptos de transformación y dependencia entre nodos, logrando una comprensión mucho más profunda de cómo se estructuran los objetos complejos en un entorno 3D.

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